A32: The 35-component Weil representation and ρ_7: exact intertwiners and an explicit map

For: Arvin Hampton · 2026-10-22, about 10:25 AM CDT.

Folder: `reports/s_family_paths/A32_35_to_rho7/` - Scripts: `exact_hom.py`, `run.py`, `numeric.py`, `exact_verify.py`, `controls.py`. - Results: `results.json`, `numeric.json`, `exact_verify.json`, `controls.json`, `X_chi3.npy`.

Ground rules - `A/scripts/weil_family.py` and the RTA_prereg runner were imported read-only. The certifier was not touched (CORE_FREEZE), and H_s was not redefined. - The cert queue was not touched. - New files only. Nothing was pushed or sent. - No TPC or RH claims. - **Cat A** = proved, or exact relative to named imports. **Cat B** = numerical or interpretive.

Verdict

1. **Exact: no equivariant map without a twist.** Hom_{$SL_2(Z)$}(ρ_{35}, ρ_7|W) = 0, and the same holds for the dual ρ_{35}, *in both directions and against ρ_7|W*. This is proved by a mod-p rank certificate at two primes (Cat A, computer certificate). It is **not** explained by disjoint T-spectra: all 12 T-eigenvalues of ρ_{35} occur in ρ_7|W. The obstruction is the S-matrix.
2. **Exact: an explicit map after the order-4 character twist χ^3.**
3. **Twist:** χ is the generator of Hom($SL_2(Z)$, C^×) ≅ Z/12, with χ(T) = e(1/12) and χ(S) = −i. So χ^3(T) = i and χ^3(S) = i.
4. **Hom dimension:** dim Hom($\rho_{35}\otimes\chi^3$, ρ_7|W) = 1.
5. **The map Φ:** it is explicit and monomial, with rank 6 and image exactly the cusp part C. It is verified exactly in Z[ζ_29400] on S and T (Cat A).
6. **Identification: C ≅ $(\rho_{35})_{(-,-)} \otimes \chi^3$**, where $(\rho_{35})_{(-,-)} = \rho_5^- \otimes \rho_7^-$ is the 6-dimensional part that is odd under both local reflections Z_5 and Z_7.
7. **Uniqueness:** this is the only one of the 12 characters that gives a non-zero Hom, and it does so with multiplicity 1. The other non-zero cases are the dual and adjoint forms of the same statement.
8. **Revised ledger entry 61.** The untwisted statement is upgraded from floating point to **exact Cat A (computer certificate)**. The conclusion "no equivariant map" holds only untwisted. Up to the χ^3 twist, C is canonically a sector of the 35-component Weil representation.

1. What the objects are (sources: RTA_prereg Amendments 1–2, RTA_implications §2, A18 §4)

- **Lattice.** A_7 = [[2,7],[7,7]], with det −35. It is an **odd** lattice.
- Amendment 1 (Cat A): there is no even binary lattice of det −35, because $b^2 - 4ac = 35 \equiv 3 \bmod 4$ is impossible.
- **D(A_7) and its form.** D(A_7) = L′/L has order 35. Since 35 is odd, b(μ,ν) = $\mu^T A^{-1}\nu$ mod 1 has a unique quadratic refinement q(μ) = $2^{-1}\cdot$b(μ,μ), with 2^{-1} = 18 mod 35.
- **The 35-dim representation ρ_{35}** is the Weil representation of (D, q):
- T e_μ = e(q(μ)) e_μ;
- S e_μ = (ε/√35) Σ_ν e(−b(μ,ν)) e_ν, with **ε = i**.
- ε is fixed by the SL_2 relations. It is the unique choice at both primes used below, and numerically it is also unique.
- **Group and level.** ρ_{35} is a representation of $SL_2(Z)$ ($S^4 = 1$, $(ST)^3 = S^2$). It needs no metaplectic cover. T has exact order 35, so the representation factors through $SL_2(Z/35)$.
- **The "35-component law" (Residual-Law B3).** Its source marks it "Not attempted". **No B3 law vector exists anywhere in the repo** (Amendment 2; A18 §4). So the representation space is the only exact object available, and this report works with it.
- **ρ_7.** This is the s-family representation on the 140 labels (`family_s.py` / `WFam(7)`), restricted to the 42-dimensional subspace W fixed by γ_7 and −1.
- It is an $SL_2(Z)$ representation, with T of **exact order 140** on W (computed here exactly; not the "level 70" sometimes quoted).
- W = E ⊕ C, where E is 36-dimensional and C is 6-dimensional.
- dim End(ρ_7|W) = 2 exactly, and dim End(ρ_{35}) = 4 exactly.
- **Matrices.**
- ρ_{35}: S and T as above. The labels are listed in `exact_verify.json` (stored as r = adj(A)·m mod 35, so μ = r/(−35)), with their q values.
- ρ_7|W: the WFam S_row and T_mono rows restricted to W orbit vectors. All phases lie in e(Z/9800), i.e. are 9800th roots of unity, and the normalisation is 1/√35.

2. Exact intertwiner computation

- **Method.** Solve $XS_1 = S_2X$ and $XT_1 = T_2X$ in exact arithmetic over F_p, with p ≡ 1 mod 29 400 (= lcm(9800, 140, 12, 8)), and with ζ ↦ a primitive 29 400th root of unity in F_p.
- √35 and ε are fixed in F_p by the relations: exactly one root of 35 and one ε satisfy them.
- Both S's carry the same factor 1/√35, so it cancels.
- **Why the mod-p result is a proof of vanishing.** The equations have coefficients in Z[ζ][1/35]. Reduction mod a prime above p is a ring map, so the rank can only drop, and **dim Hom over C ≤ dim over F_p**. A mod-p nullity of 0 therefore proves Hom = 0 in characteristic 0.
- Two primes were used, p = 29 401 and 14 729 401, with identical results.
- **Scope of the check.** 96 Hom spaces in total: {ρ_{35}, ρ_{35}} ⊗ χ^k ($k = 0..11$) → {ρ_7|W, ρ_7|W}, in both directions.
- **T-spectra (exact).**

- ρ_{35} has eigenvalues e(t), with multiplicities:

t	0	1/7	2/7	4/7	2/5	3/5	6/35	19/35	24/35	26/35	31/35	34/35
mult.	1	2	2	2	2	2	4	4	4	4	4	4

- ρ_7|W has 36 distinct eigenvalues of order dividing 140; the full list is in `results.json`.

- **Overlap, untwisted:** all 12 eigenvalues of ρ_{35} occur in ρ_7|W, each with multiplicity 1 there. So the spectrum allows up to 37 matched unknowns, and the vanishing comes from S.
- **Overlap after χ^3:** the shifted values t + 1/4 all occur too.
- **Results.**

Variant	dim Hom (exact upper bound, two primes)	Status
$\rho_{35} \to \rho_7\|W$, and the reverse direction	0	**Cat A, = 0**
$\rho_{35}^* \to \rho_7\|W$, and the reverse direction	0	**Cat A, = 0**
$\rho_{35}, \rho_{35}^* \to \rho_7\|W^*$	0	**Cat A, = 0**
$\rho_{35} \otimes \chi^k \to \rho_7\|W$, for $k \neq 3$	0	**Cat A, = 0**
$\rho_{35} \otimes \chi^3 \leftrightarrow \rho_7\|W$	**1**	**Cat A, = 1** (lower bound by the explicit map in §3)
$\rho_{35} \otimes \chi^9 \leftrightarrow \rho_7\|W$	1	the complex-conjugate statement
all other combinations	0	Cat A

- **Consequences for subspaces.** For the subspaces E and C, every Hom vanishes except $\chi^3 \to C$, which has dimension 1. This follows because a Hom into a subrep embeds in the Hom into W, and the image in the χ^3 case is C (§3).
- **Metaplectic (half-integral η-power) twists.** A genuine character of $Mp_2(Z)$ acts by −1 on the central element S^4. Both representations here are trivial on S^4, so all those Hom spaces are 0 (Cat A, one line).
- **Weight-shift twists by η^{2k}.** These are exactly the χ^k above, and the list is exhaustive: χ generates all 12 characters of $SL_2(Z)$.

3. The explicit map (Cat A)

- **Construction.** Φ: $\rho_{35} \otimes \chi^3 \to \rho_7|W$ was found in floating point.
- Its residual is $1.4 \cdot 10^{-16}$, and the gap to the next singular value is 1.05.
- All 24 non-zero entries have equal modulus, and their phases are exact multiples of 1/29 400.
- **Exact verification.** Rescaled to unit entries, Y was verified **exactly** in the group ring Z[Z/29400], reduced modulo the cyclotomic polynomial Φ_{29400}:
- $Y \cdot S_{35}' = S_7 \cdot Y$: **210 of 210 entries agree**;
- $Y \cdot T_{35}' = T_7 \cdot Y$: **24 of 24 entries agree**.
- The check uses the unnormalised orbit basis. All six target orbits have size 4, so the normalisation is a single scalar.
- Generators suffice, so Φ is an $SL_2(Z)$-intertwiner.
- **Rank and image.** Rank 6: the six image rows have disjoint column supports.
- The image is spanned by the W-orbit basis vectors with indices 26, 28, 30, 32, 34 and 36.
- That image is C, because W = E ⊕ C with End = 2, so E and C are the only non-zero proper subreps, and C is the 6-dimensional one.
- **So C is spanned exactly by six W-orbit vectors.**
- **Kernel (29-dim).** It contains every e_μ with μ in the 5-torsion or the 7-torsion subgroup. These are the 11 labels with $q \in \{0, 1/7, 2/7, 4/7, 2/5, 3/5\}$, i.e. with a zero local component. The remaining 18 dimensions of the kernel come from the other local-sign sectors on the 24 labels.
- **What Φ does to the 35 components.** The 24 "primitive" labels, those with both local parts non-zero ($q \in (1/35)Z$ with denominator 35), map 4-to-1 onto the six C basis vectors. Each row is an orbit $\{\pm\mu, \pm\mu'\}$, where μ′ is μ with one local part negated:

C row	labels (coefficient)	q
26	[7,23] (+1), [28,12] (+1), [7,33] (−1), [28,2] (−1)	34/35
28	[7,18] (+1), [28,17] (+1), [7,3] (−1), [28,32] (−1)	24/35
30	[7,13] (+1), [28,22] (+1), [7,8] (−1), [28,27] (−1)	19/35
32	[14,11], [21,24] (e(1/5)); [14,31], [21,4] (e(7/10))	31/35
34	[14,6], [21,29] (e(1/5)); [14,1], [21,34] (e(7/10))	26/35
36	[14,26], [21,9] (e(1/5)); [14,16], [21,19] (e(7/10))	6/35

- **Reading the table.** Each row is even under μ ↦ −μ and odd under the local reflection. So the source of Φ is the ($Z_5 = -$, $Z_7 = -$) sector $\rho_5^- \otimes \rho_7^-$, of dimension 2·3 = 6. This is consistent with A ("for odd s, C lies in the ($Z_p = -1$)-sectors").
- **Is Φ canonical?** Yes, up to a scalar: Hom is 1-dimensional and both sides are irreducible. It is unique up to scaling once χ^3 is fixed.
- **Applying it to the B3 law.** Φ would send any 35-component vector to its (−,−) primitive sector. This **could transport a B3 law vector into C**, but no such vector exists, so nothing is transported here.

4. Controls and interpretation (Cat B)

- **Floating-point controls** (`controls.json`, χ^k for k = 0..11): the same single χ^3 hit (or its dual) appears for −[[2,1],[1,18]], the dual of [[2,1],[1,18]], [[5,0],[0,7]], [[3,1],[1,12]] and [[2,1],[1,−17]]. It does **not** appear for ±[[6,1],[1,6]], [[1,0],[0,35]] or [[4,1],[1,9]].
- **So the map is not specific to A_7.** It exists whenever the local odd parts of the order-35 discriminant form match those of D(A_7) (Cat B: controls in floating point only).
- **Interpretation (Cat B).** χ^3 is the multiplier of η^6 (weight 3). One reading is that C is the doubly-odd sector of the D(A_7) Weil representation, shifted by a weight-3 multiplier. Whether the Residual-Law B3 law, if ever built, lives in this sector is open.

5. Status

- **Cat A**

- The untwisted Hom = 0 results (exact, CC at two primes; one prime suffices).
- The 12-character exhaustive table.
- The explicit Φ with its exact Z[ζ] verification, its rank 6 and image C, its kernel, and the component table.
- $C \cong \rho_5^- \otimes \rho_7^- \otimes \chi^3$.
- **Inherited imports:**
  - ε = i, from the SL_2 relations (unique mod p; standard Weil-representation theory);
  - WFam's definitions of ρ_7 and W;
  - E being a subrep (RTA_implications), with End = 2 exact by the same mod-p bound plus E, C ≠ 0.
- **Cat B:** the controls, and the physical or "law" reading.
- **Open:** the B3 law vector, which has never been built.
- **Ledger entry 61 (revised):**
- "Hom(ρ_{35}, ρ_7|W) = Hom(ρ_{35}*, ρ_7|W) = 0" is now **exact (CC)**.
- **Amendment:** "after the χ^3 twist there is a unique (up to scalar) intertwiner Φ, with image $C \cong (\rho_{35})_\{(-,-)\} \otimes \chi^3$, verified exactly". This moves from the negative section to section 3 (Bridge C) as a positive Cat A result.